

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 - Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 - Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	2 ND YEAR	Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Visualization-Based Technical Presentation on Growth Functions

Analyze growth functions $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$ using visualization tools in a technical presentation format. Clearly define the purpose of comparing these functions for understanding algorithm efficiency. Use graphs, tables, or charts to show how each function behaves as the input size increases. Explain which functions remain practical for large-scale problems and justify your reasoning. Present the conclusions clearly and professionally by linking growth behavior to algorithm selection.

TECHNICAL PRESENTATION · ALGORITHM ANALYSIS

Analyzing Growth Functions

Comparing $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$ and $O(2^n)$

$O(\log n)$

$O(n)$

$O(n \log n)$

$O(n^2)$

$O(2^n)$

What Is a Growth Function?

A growth function $f(n)$ describes how the number of operations an algorithm performs increases as the input size n grows. Big-O notation, $O(f(n))$, gives the upper bound of this growth — letting us compare algorithms by their long-term efficiency rather than raw execution time.

Formal Definition

$f(n) = O(g(n))$ if there exist constants $c > 0$ and n_0 such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$

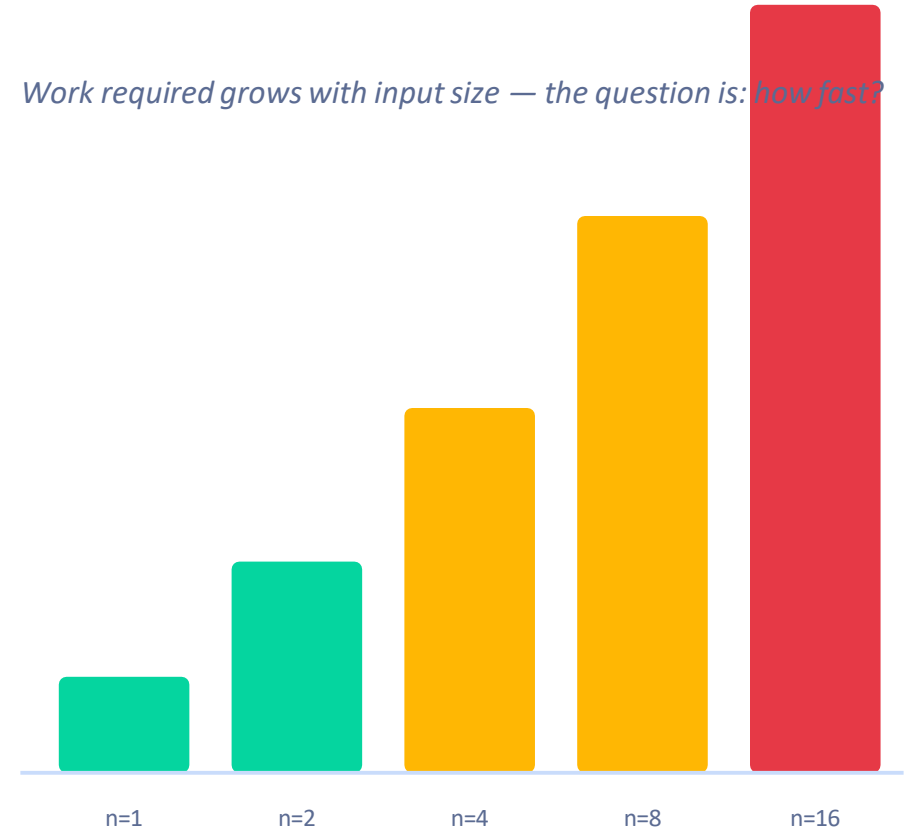
GROWTH HIERARCHY (SLOWEST → FASTEST GROWING)



Why Compare Growth Functions?

Two algorithms can solve the same problem correctly yet behave very differently as data grows. Runtime in seconds is machine-dependent and unreliable — we need a way to describe how work scales purely with input size, independent of hardware.

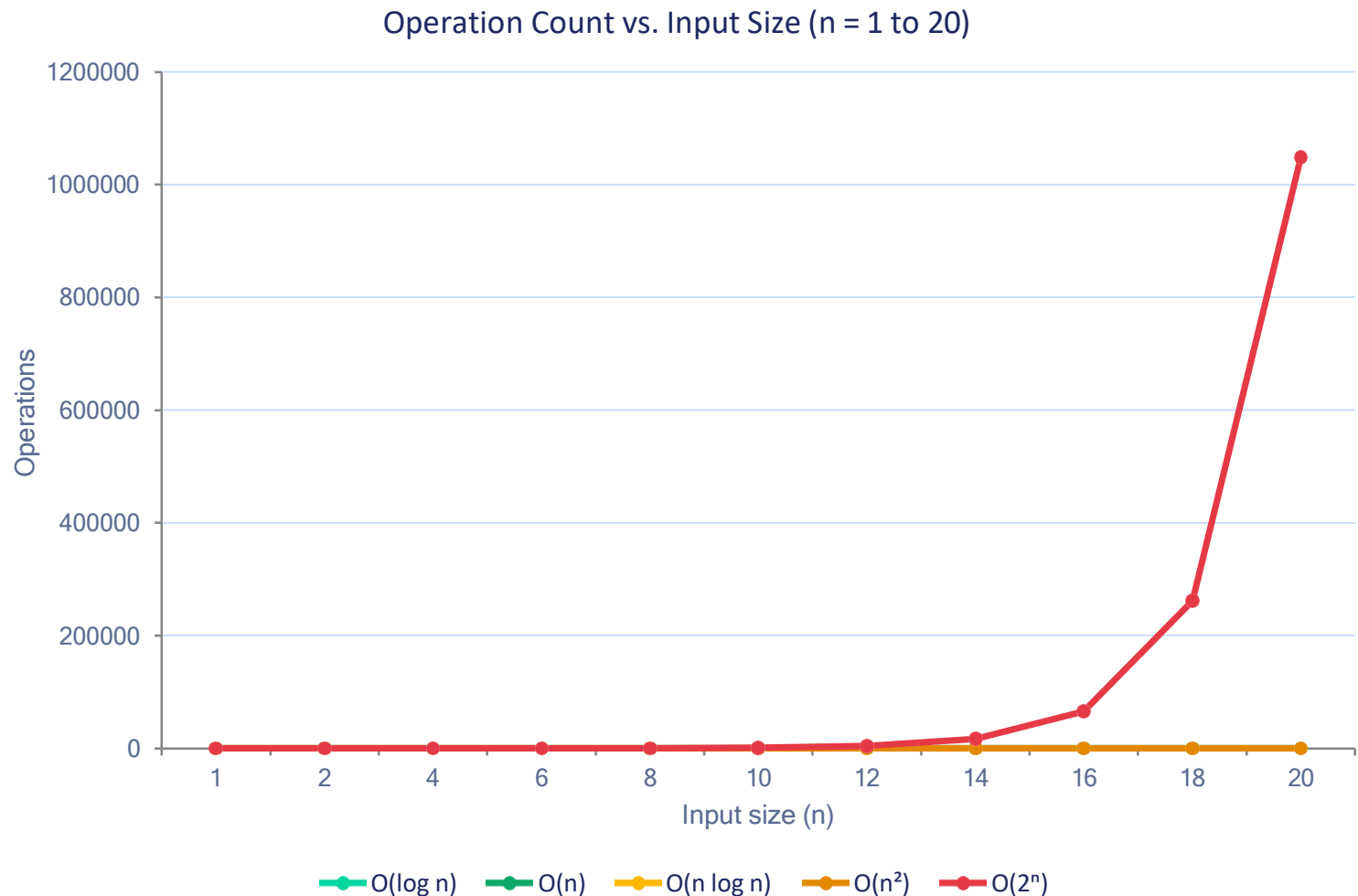
- Predict algorithm behavior before touching large real-world data
- Compare algorithms independently of hardware or language
- Identify bottlenecks early and choose the right approach
- Justify design decisions with evidence, not guesswork



Five Core Growth Functions

Notation	Class	Behavior	Example Algorithm
$O(\log n)$	Logarithmic	Halves the problem each step	Binary search
$O(n)$	Linear	Touches every element once	Linear search
$O(n \log n)$	Linearithmic	Divide-and-conquer with merge/scan	Merge sort, quicksort
$O(n^2)$	Quadratic	Compares every pair of elements	Bubble sort, insertion sort
$O(2^n)$	Exponential	Doubles work with every extra input	Subset generation, brute-force search

Visualizing Growth Behavior



Key Observation

At $n = 20$, $O(2^n)$ exceeds 1 million operations while $O(n^2)$ reaches only 400.

Exponential growth overwhelms the chart almost immediately — even at small input sizes.

$O(\log n)$ and $O(n)$ stay nearly flat, confirming why they scale best.

$2^{20} = 1,048,576 \text{ ops}$

Linking Growth Rate to Algorithm Choice



$O(\log n)$

Search engines & databases

Binary search trees / B-trees keep lookups fast over billions of records



$O(n \log n)$

General-purpose sorting

Merge sort / quicksort chosen as the practical standard for large data



$O(n^2)$

Small-scale problems

Insertion sort still wins for tiny or nearly-sorted arrays



$O(2^n)$

Only small n / brute force

Replaced with heuristics or dynamic programming as n grows

CONCLUSION

Growth Behavior Should Drive Algorithm Choice

1

Logarithmic and linear functions scale best

They stay fast even for millions of inputs — always prefer them when possible.

2

$n \log n$ is the practical ceiling for sorting

It balances speed and simplicity for most large-scale algorithms.

3

Quadratic is a caution, exponential is a wall

$O(n^2)$ suits small inputs only; $O(2^n)$ must be avoided or replaced at scale.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

